

Flight Rank

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[FlightRank 2025: Aeroclub RecSys Cup | Kaggle](#)

kaggle



How Many of you uses YouTube, Facebook, Booking, etc.. Apps?

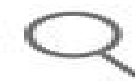


Booking



How many of you uses the search bar?

Search...



How many of you find what they are searching for?

Our Advice
Buy now

Prices are unlikely to decrease within 7 days ⓘ

Track prices ☐ Off

1009 of 1620 flights

Recommended filters

☐ Hide 2+ stops

Stops

☒ Nonstop \$183

☒ 1 stop \$195

☒ 2+ stops \$271

Fee Assistant ⓘ

Cheapest
\$183 • 3h 17m

Featured ⓘ

\$183 • 3h 17m

Quickest
\$183 • 3h 17m

Other sort

eDreams Find the best deals on eDreams
Compare flights from more than 600 airlines

☐	spirit 6:00 am – 9:20 am Spirit Airlines	nonstop	3h 20m BOS - FLL	🔒 0 🛒 0 \$183 Economy eDreams View Deal
☐	spirit 7:36 pm – 10:51 pm Spirit Airlines	nonstop	3h 15m FLL - BOS	

Ad

Best

Cheapest

☐	spirit 6:00 am – 9:20 am Spirit Airlines	nonstop	3h 20m BOS - FLL	🔒 0 🛒 0 \$183 Economy eDreams View Deal
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Problem Definition

- Build a recommendation system that predicts which flight option a business traveler is most likely to choose within a given search session.
- The model has to rank all the available flight options from best to worst based on the user's specific preferences, company policies and other factors.



Motivation

- Our goal is to create a model that predicts future airline recommendation system's performance using a large and complex dataset from a global competition.
- Learn how to deal with real-world problem.





Methodology



Data

- Each search session = ranker_id → multiple flight options
- Train: selected (1 = chosen, 0 = not chosen)
- Test: predict rank (1= Best, N = worst)
- train.parquet → Training data
- Test.parquet → Test data (no target)



Data

- User Info: gender, nationality, frequent flyer, VIP
- Company Policy: tariff code, policy compliance
- Route & Search: route, request date/time
- Pricing: total price, taxes
- Flight Timing: outbound/return times & durations
- Segments: airline, aircraft, baggage, seats, cabin class
- Rules: cancellation & exchange penalties



Data Analysis

- Drop columns with null values $> 98\%$
- Drop unnecessary columns.
- Convert data type columns from string into date & time using regex and polar functions.
- Total Price



Model Selection

XGBoost

1. Performance (Accuracy / Ranking Score)

- 38%

2. Training & Inference Speed

- Fast training, but can be slower with large categorical data (needs one hot encoding).

3. Handling of Categorical Features

- Needs preprocessing (label encoding or one-hot).

4. Robustness & Ease of Use

- Requires more manual hyper parameter tuning to reach good performance.

CatBoost

1. Performance (Accuracy / Ranking Score)

- 40%

2. Training & Inference Speed

- Automatically handles categorical features efficiently, usually faster on inference when many categorical features exist.

3. Handling of Categorical Features

- Native categorical handling no need for heavy preprocessing.

4. Robustness & Ease of Use

- Often works well out of the box with fewer parameter.



Challenges

1. Data size = 5.27 GB
2. CPU & TPU VM v3-8 limitation
3. Selecting the best model



Demo



Result

FlightRank 2025: Aeroclub RecSys Cup							Late Submission	...
Overview	Data	Code	Models	Discussion	Leaderboard	Rules	Team	Submissions
#	Team	Members	Score	Entries	Last	Solution		
1	gezi		0.54062	100	3d			
2	Phaedrus		0.53548	51	3d			
3	AF	 	0.53548	190	3d			
4	mango789 & Ionut Visan	 	0.53438	210	3d			
5	Taos David		0.52970	56	3d			
6	Mikhail Golubchik		0.52814	105	3d			
7	kif		0.52795	63	4d			
8	XS 330		0.52529	39	3d			
9	Muhammad Hassaan		0.52446	156	3d			
10	Dmitriy Ch		0.52437	134	4d			



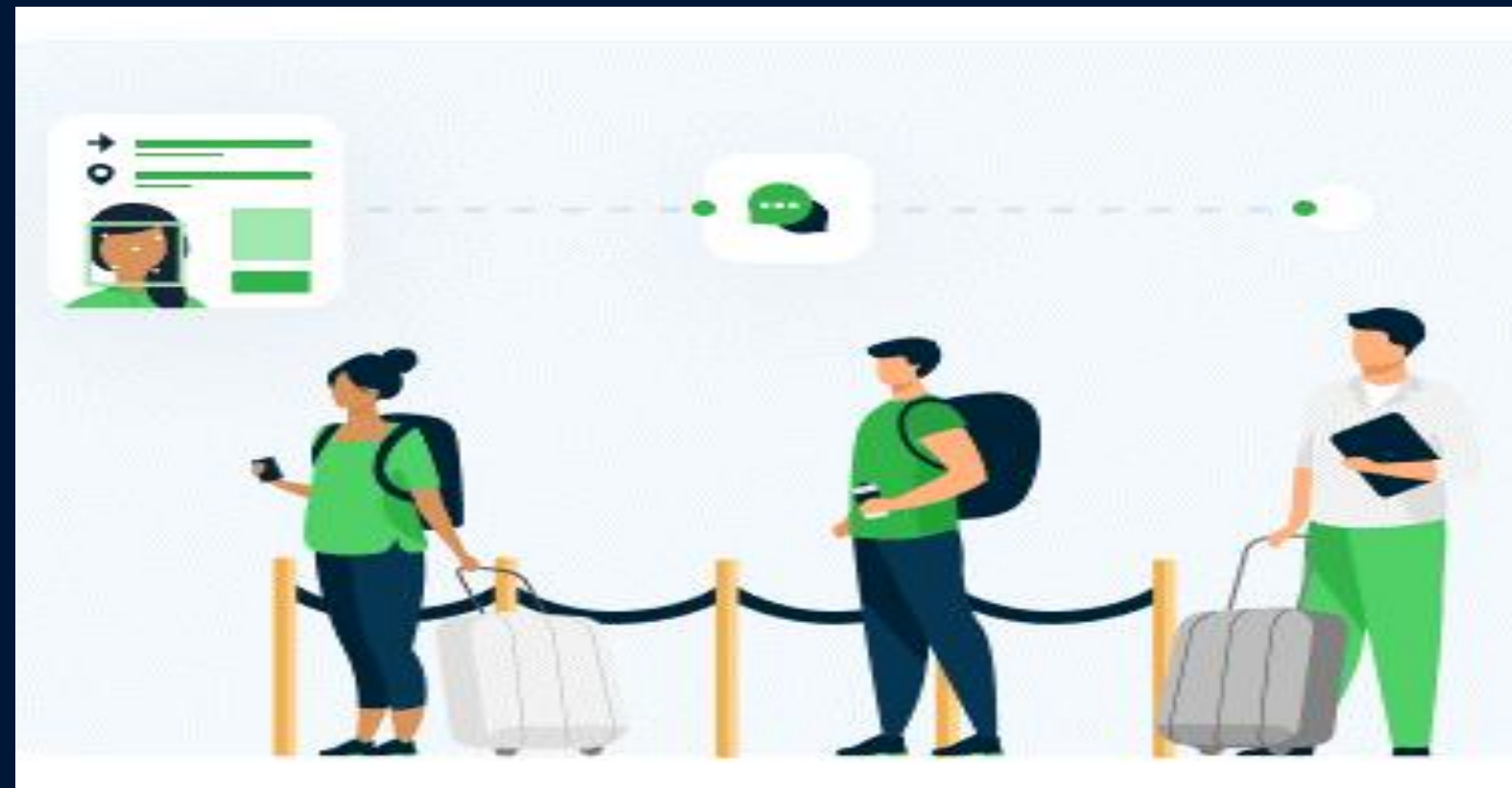
Result

FlightRank 2025: Aeroclub RecSys Cup							Late Submission		...
Overview	Data	Code	Models	Discussion	Leaderboard	Rules	Team	Submissions	
436	—	gemini				0.40000	1	1mo	
437	▼ 2	Sagor Kumar Mitra				0.40540	2	3d	
438	▼ 4	Dayou Wang				0.40459	1	7d	
439	▼ 2	Jay Prajapati				0.40451	1	1mo	
440	—	Zhong Haitian				0.40354	7	2mo	
441	—	akmonymous				0.40350	2	1mo	
442	—	sahar mah3r99				0.40226	10	2d	



Future Work

Build a mobile application for the system.





THANK
YOU!